Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Engineer, NoSQL and In-Memory Database** role with **Amazon Web Services (AWS)**. I bring over three years of professional experience building distributed systems and backend services, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY with a 4.0 GPA.

Through my experience across two engineering roles, I have built production-grade distributed systems that directly reflect the sub-millisecond latency, fault-tolerant in-memory architecture, and high-availability engineering this position demands. As a Software Engineer at Sperse, I engineered a high-throughput distributed systems service exposing 160+ tools over a NoSQL-backed multi-tenant backend — using asynchronous messaging and event-based routing with OpenTelemetry instrumentation for real-time failure detection. To maintain state across concurrent agent interactions, I implemented Redis-backed in-memory NoSQL and SQLite-checkpointed multi-turn state management with high-availability persistence, mirroring the core challenges of sub-millisecond in-memory database design at scale. I also built auto-remediation pipelines with real-time monitors that detect and contain unsafe behaviors before release — the same operational rigor that AWS's in-memory services demand. Prior to Sperse, as a Software Engineer at Qualitest, I orchestrated long-running distributed jobs with AWS Step Functions and Lambda, implementing event-based workflows with fault-tolerant error recovery and auto-remediation across 100K+ weekly executions. I also applied low-level optimizations to PostgreSQL — window functions, materialized views, and composite indexes — achieving sub-millisecond read latency and high-throughput data access under production load. Together, these roles have prepared me to contribute to AWS's NoSQL and in-memory database infrastructure — building highly available, scalable Redis-backed services used by millions of customers.

My academic research experience deepened my understanding of distributed evaluation systems and systematic fault detection, skills that translate directly into the operational excellence and customer-centric engineering discipline this team values. As an AI Researcher at the Pace Artificial Intelligence Lab, I built a distributed agent-evaluation pipeline ensembling LLM judges and reward models with asynchronous parallel scoring across compute nodes — and implemented a LangGraph eval pipeline that increased tool-call success by 50% through iterative fault detection and feedback loops. Beyond building these systems, I have consistently taken ownership of mentoring junior contributors, explaining complex technical tradeoffs clearly — directly reflecting the mentorship responsibilities central to this role and AWS's culture of raising the performance bar as Earth's Best Employer.

I have always been passionate about **customer obsession** and **passion for invention** — the twin principles at the heart of Amazon's leadership culture — and envision myself contributing to AWS's mission to build the world's most comprehensive and broadly adopted cloud platform. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan